MIL-C-36525
10 May 1967

MILITARY SPECIFICATION

CUFF, ENDOTRACHEAL TUBE

1. SCOPE

1.1 Scope. This specification covers endotracheal tube cuffs suitable for occluding the lumen of the human trachea when used with endotracheal tubes in a positive pressure anesthetic procedure.

1.2 Classification. Cuffs shall be of the following sizes, as specified (see 6.1):

Size 1 - 6.5 mm id
Size 2 - 8.0 mm id
Size 3 - 9.5 mm id

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Box, Setup.

Military

MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

FSC 6515

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STANDARDS

Federal

FED-STD-601 Rubber, Sampling and Testing

Military

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 Marking for Shipment and Storage.

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publication. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

United States of America Standards Institute

USA Z79.1-1960 American Standard Specifications for Anesthetic Equipment: Endotracheal Tubes.

(Application for copies shall be addressed to the United States of America Standards Institute, 10 East 40th Street, New York, New York 10016.)

3. REQUIREMENTS

3.1 Material. Material shall be natural latex of the highest quality throughout, and suitable in all respects for the purpose intended.

3.2 Style, design, and construction. Shall be a double-wall, bulge slip-on type, Guedel design endotracheal tube cuff. Cuffs shall inflate evenly and shall be suitable for occluding the lumen of the human trachea when used with endotracheal tubes in a positive-pressure anesthetic procedure. Shall be in accordance with applicable portions of paragraphs 4 and 5, for size indicated, of United States of America Standards Institute (USA) Standard Z79.1-1960. The cuff size and the size of endotracheal tubes with which each cuff shall be suitable for use, shall be as indicated below:

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Cuff size	I.D. cuff in mm.	For use with endotracheal tubes
1	6.5	5-6.5 mm. I.D.
2	8.0	7-8.5 mm. I.D.
3	9.5	9-10 mm. I.D.

3.3 Sterilization. When tested as specified in 4.4.1, cuffs shall show no signs of deterioration.

3.4 Physical Properties. The minimum physical properties of the cuffs before aging shall be as follows:

Tensile strength 3500 lbs. per square inch
Ultimate elongation 800 percent
Tensile and elongation tests shall be conducted in accordance with 4.4.2.

3.5 Resistance to aging. The tensile strength and ultimate elongation of the cuffs shall not decrease more than 25 percent of the value before aging nor shall the rubber latex become tacky or brittle. Resistance to aging test shall be conducted in accordance with 4.4.3.

3.6 Air leakage. The cuffs shall show no leakage of air when tested as specified in 4.4.4.

3.7 Workmanship. Workmanship shall be first class throughout. Cuffs shall be free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspections. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties).

4.1.2 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105.

4.2.2 For tests. Sampling for each test shall be conducted in accordance with MIL-STD-105, inspection level S1, except that the acceptance number shall be zero for all sample sizes.

4.2.3 Test sample. Where possible, the same sample unit shall be used for the determination of two or more test characteristics.

4.3 Examination. The cuffs shall be examined to determine compliance with all requirements contained in this specification.

4.4 Tests. Tests shall be conducted as follows, to determine compliance with specification requirements.

4.4.1 Sterilization.

4.4.1.1 Steam sterilization. Cuffs shall be steam autoclaved for 20 minutes at 250° Fahrenheit (F.) and 15 pounds per square inch gage, then allowed to cool to room temperature (68° F. to 72° F.).

4.4.1.2 Boil sterilization. Cuffs shall be boiled in distilled water for 30 minutes and then allowed to cool to room temperature (68° F. to 72° F.).

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4.4.1.3 Cold sterilization. Cuffs shall be immersed in a chemical disinfectant for 20 minutes at room temperature (68° F. to 72° F.), followed by five water rinses.

4.4.2 Physical properties. Tensile strength and ultimate elongation tests shall be performed in accordance with methods 4111 and 4121 respectively in Federal Test Method Standard 601.

4.4.3 Resistance to aging. The accelerated aging test shall be performed in accordance with method 7221 in Federal Test Method Standard 601.

4.4.4 Air leakage. The cuff, after being submitted to the sterilization tests and aging test , shall be slipped over an endotracheal tube. (See 3.2 for correlation of cuff size to size of endotracheal tube.) The cuff shall be inflated with 25cc of air, using a syringe or other suitable method to measure the air, and the tube connected to the cuff shall be clamped at the end to retain the air. The specimen shall be completely submerged for 5 minutes ± 5 seconds in a container of water suitable to permit good visual observation of the specimen. The inflated cuff shall be examined for leakage of air, and other evidence of failure.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One cuff, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in table I shall be packaged in each unit, intermediate, protective, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table I. Packaging Quantities

Unit package	Intermediate package	Protective package	Exterior container
1 unit	12 units	144 units	1152 units

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5.1.2.1 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Unit packaging. Tubing shall be formed into a coil of such diameter that it will not kink or be permanently deformed. Each unit so prepared shall be packaged in an envelope or carton of suitable size and design. Envelope or carton shall be adequately secured to prevent accidental opening.

5.1.3.2 Intermediate package. Intermediate package shall be a carton of appropriate size constructed in accordance with PPP-B-566 or PPP-B-676, except that commercial colors shall be acceptable. Closure shall be adequate to prevent spillage of contents under normal handling.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for type 2 load, and constructed in accordance with PPP-B-585, style 3, for class 3 use; PPP-B-601, table II, using type III, class 1 plywood; PPP-B-621, table III; or PPP-B-636, type CF, class weather-resistant.

5.2.1.2 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes.

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5.2.1.3 Closure. Closure of wood boxes shall be in accordance with appendix of applicable box specification. Closure of each fiberboard box shall be as specified in the appendix of PPP-B-636. In addition, waterproofing shall be in accordance with the appendix.

5.2.1.4 Strapping. Strapping, when required, shall be in accordance with appendix of applicable box specification.

NOTE: Strapping shall not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall conform to PPP-B-601, table I, using type III, class 1 plywood, or shall be designed for type 2 load, and constructed in accordance with PPP-B-585, style 3, for class 2 use; or PPP-B-621, table I, or PPP-B-636, type CF, class domestic. Closure of fiberboard box shall be in accordance with method I of the box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Interior packages. Each unit package, intermediate package, and protective package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use. Cuff is intended for use in occlusion of human trachea when used with endotracheal tubes in positive pressure anesthetic procedure.

6.2 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.
(b) Size of cuff.
(c) Selection of applicable levels of packing (see 5.1 and 5.2).

6.3 This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet military requirements.

Custodians:
Army - MD
Navy - MS
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Review activities:
Army - MD
Navy - MS
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Preparing activity:
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CAUTION

NOTICE TO BIDDERS/OFFERORS

DO NOT CONDITION OR BASE YOUR BID/OFFER ON ANY CURRENT PROCUREMENT ON THE INFORMATION SUBMITTED ON THIS FORM SINCE ANY CHANGES OR DELETIONS IN THE SPECIFICATIONS MAY RENDER YOUR BID/OFFER NON-RESPONSIVE IN WHICH CASE IT CANNOT BE CONSIDERED FOR AWARD.

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